

Lecture 21: C.I. Examples, Asymptotic/Approximate C.I.

Ex 3 $X_1, \dots, X_n$ are iid observations from $\text{Exp}(\theta)$, with pdf $f_\theta(x) = \frac{1}{\theta} e^{-x/\theta}$, $x > 0$, $100(1-\alpha)\%$ C.I. for θ , $\theta > 0$.

For $i=1, \dots, n$, $Y_i = \frac{X_i}{\theta} \sim \text{Exp}(1)$, so that $\sum_{i=1}^n Y_i = \frac{\sum_{i=1}^n X_i}{\theta} \sim \text{Gamma}(n, 1)$ distn can be a pivot for

Better to use $2 \sum_{i=1}^n Y_i = \frac{2 \sum_{i=1}^n X_i}{\theta} \sim \chi^2_{2n}$

Then an equal tailed $100(1-\alpha)\%$ C.I. for θ can be given using,

$$P\left[\chi^2_{1-\alpha/2, 2n} \leq \frac{2 \sum_{i=1}^n X_i}{\theta} \leq \chi^2_{\alpha/2, 2n} \right] = 1-\alpha$$

The $100(1-\alpha)\%$ C.I. obtained from above is $\left[\frac{2n\bar{X}}{\chi^2_{\alpha/2, 2n}}, \frac{2n\bar{X}}{\chi^2_{1-\alpha/2, 2n}} \right]$.

Ex 4 $X_1, \dots, X_n$ are iid observations from $U(0, \theta)$, $\theta > 0$

$100(1-\alpha)\%$ C.I. for θ

Here, also $\forall i=1, \dots, n$, $Y_i = \frac{X_i}{\theta} \sim U(0, 1)$.

Then $Y = \frac{X_{(n)}}{\theta}$ can be a pivot with pdf

$$f_\theta(y) = ny^{n-1}, \quad 0 < y < 1$$

($Y \sim \text{Beta}(n, 1)$)

Include the max value

$$P[k_1 \leq Y \leq 1] = 1-\alpha$$

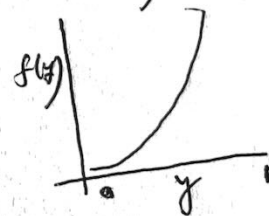
$$\Rightarrow k_1 = \alpha^{1/n}$$

$$\Rightarrow P[X_{(n)} \leq \theta \leq X_{(n)} \alpha^{-1/n}] = 1-\alpha$$

(Equivalently $P[X_{(n)} \leq \theta \leq c X_{(n)}] = 1-\alpha$)

$\therefore X_{(n)}$ is a natural l.b. for θ in this case.

Thus a $100(1-\alpha)\%$ for θ is $[X_{(n)}, X_{(n)} \alpha^{-1/n}]$.



In addition to, (Exact) C.I.s we have the Asymptotic / Approximate C.I.s

1. Direct use of the (Lindberg-Levy) CLT:

$\{X_n\}_{n \geq 1}$ seq. of iid r.v.s with $E(X_i) = \mu$ and $\text{Var}(X_i) = \sigma^2 < \infty$

Let $S_n = \sum_{i=1}^n X_i$, then

$$\frac{S_n - E(S_n)}{\sqrt{\text{Var}(S_n)}} \stackrel{L}{\sim} Z \sim N(0, 1)$$

which implies,

$$\frac{\bar{X}_n - \mu}{\sigma/\sqrt{n}} \stackrel{L}{\sim} Z \sim N(0, 1)$$

so that, $\sqrt{n}(\bar{X}_n - \mu) \xrightarrow{L} N(0, \sigma^2)$ as $n \rightarrow \infty$

Using Δ -method for any $g(\mu)$,

$$\sqrt{n}(g(\bar{X}_n) - g(\mu)) \xrightarrow{L} N(0, (g'(\mu))^2 \sigma^2), \text{ as } n \rightarrow \infty$$

(For application w/ non-normal skewed dists, like $\text{Exp}(\lambda)$, $\text{Gamma}(\alpha, \beta)$, $\text{Binomial}(n, p)$, $\text{Poisson}(\lambda)$ etc. then normal approx is possible by using CLT).

e.g., $X_1, \dots, X_n$ iid obs from $\text{Exp}(\lambda)$, $\lambda > 0$ dist.

$$\text{Then, } \sqrt{n}(\bar{X}_n - \theta) \xrightarrow{L} N(0, \theta^2)$$

$$\text{and } \sqrt{n}\left(\frac{1}{\bar{X}_n} - \frac{1}{\theta}\right) \xrightarrow{L} N(0, 1/\theta^2)$$

Like the exact C-I case with normal dist, use, e.g., $\pm t_{\alpha/2}$ in the confidence limits. The resultant C-I. is an asymptotic/approximate C-I.

2. Use of asymptotic normality of MLE

$\hat{\theta}_{MLE}$ is mle of θ , then

$$\sqrt{n}(\hat{\theta} - \theta) \xrightarrow{L} N(0, \frac{1}{I(\theta)})$$

$$\text{and } \sqrt{n}(\hat{g}(\theta) - g(\theta)) \xrightarrow{L} N(0, \frac{(g'(\theta))^2}{I(\theta)})$$

$$\text{ie, } \sqrt{n}(g(\hat{\theta}) - g(\theta)) \xrightarrow{L} N(0, \frac{(g'(\theta))^2}{I(\theta)})$$

Can be used in multiparameter case, as well.

$$\sqrt{n}(\hat{\theta}_{n \times p_1} - \theta_{p_1}) \xrightarrow{L} N_p(0, (I(\theta))^{-1})$$